

TECHNICAL SPECIFICATION FOR TENSION HARDWARE FITTINGS FOR T&C TYPE DISC INSULATOR COMPRISING OF CROSS-ARM STRAP BOLTED TYPE TENSION CLAMP TO BE USED FOR ACSR WOLF & ACSR DOG CONDUCTOR

1.0 SCOPE :

- j) The Specification covers the design, manufacture, testing (preferably at manufacturer's works before supply) and delivery of the combined unit of Cross Arm Strap bolted type Tension Clamp including loading, unloading anywhere in West Bengal.
- l) The combined units offered shall be complete with all components which are necessary (excepting disc insulator) or usual for their effective performance and easy maintenance and inter changeability at site. Such parts shall be deemed to be within the scope of contract.

2.0 GENERAL REQUIREMENTS

- i) Ball diameter should be 16 mm.
- ii) Hardware Fittings with ultimate tensile strength of 70 KN should be used for ACSR Wolf and Dog conductors.
- iii) The Fittings shall be free from defects, corrosion protected and shall meet the requirements of Galvanizing Test etc as per IS.
- iv) All forging and casting shall be of good finish and free from flaws and other defects. The edges of the fittings such as the tongue, clevis and holes shall be rounded.
- v) All parts of different fittings which provide the interconnection shall be made such that sufficient clearance is provided at the connection point to ensure free movement. All tongue and clevis connection shall be free in this manner but care shall be taken that too much clearance for the tongue & clevis is avoided.
- vi) Spring washer should be electro galvanized- Coating thickness as per IS: 1573-1986.

U bolt, Hexagonal Bolt, Nut, Plain Washer and all other ferrous parts shall be Hot dip Galvanized. In case of Hot Dip Galvanization, minimum Value of Mass of zinc coating should be 610 g/m².

The split pin to be used on the cotter pin shall be of Humpback type & shall be made of Stainless Steel conforming to IS: 5522-1992 with a minimum hardness of 160 HV.

3.0 STANDARD

The Combined units shall conform to the provisions of Indian Standards relevant to each individual component except where specified otherwise

The following I.S. amended upto date shall be the guideline for manufacture & testing.

i)	IS 2486 (Part 1) : 1993	:	Specification for metal fittings of insulators for overhead power lines with nominal voltage greater than 1000 V.
ii)	IS 2486 (Part 2) : 1989 IS : 2486 (Part III) - 1989	:	Specification for Insulator fittings for overhead power lines with nominal voltage greater than 1000 V.
iii)	IS 4759 : 1996	:	Specification for hot-dip zinc coatings on structural steel and other allied products .
iv)	IS : 6745 - 1872	:	Determination of mass of zinc coating on zinc coated iron and steel articles.
v)	IS : 2633 - 1986	:	Method for testing uniformity of coating on zinc coated.
vi)	IS:1573 - 1986	:	Specification for electroplated coatings of zinc on iron and steel.

MATERIALS AND OTHER REQUIREMENTS :

CROSS-ARM STRAPS (fig-2):

Cross-arm straps shall be manufactured from MS Flat hot dip galvanized and to connect the cross-arm/bracket of the structure at one end and the Ball Clevis at the other end.

It should be complete with three hexagonal bolts, nuts, spring washers and Cotter pin at the threaded end to lock the unit. Minimum Threaded portion of the bolt shall be **30mm**.

CROSS-ARM STRAPS to be used for	Minimum Ultimate strength (Tensile strength)
ACSR Wolf Conductor	70 KN
ACSR Dog Conductor	70 KN

Dimensions shall be in accordance with IS: 2486 (Part-2) unless otherwise specified.

Tongue and Clevis (fig-4):

All forgings shall be of good finish and free from any flaws and any other defects which may cause decrement of efficiency while in operation. Connection/attachment with other component of the unit shall provide reasonable clearance/ensure free movement at the connecting/attaching point. **Care should be taken to avoid too much clearance while used with insulators-**

Bolted type Tension Clamp (fig-3) :

All forgings & castings shall be of good finish and free from flaws or any other defects which may cause decrement of efficiency while in operation. The edges on the outside of the fittings and the grooves shall be smooth & rounded. Sharp radius of curvature, ridges etc. which may lead to localised pressure or cause damage to the conductors in service shall be avoided. The clamp shall permit the conductor to slip before the failure of conductor occurs.

Tension clamps for ACSR Wolf/Dog conductor shall be three bolted type with minimum one no. keeper. Ultimate tensile strength of Tension clamps should not be less than 70 KN .The material for Tension clamps & keeper should be high strength Aluminium Alloy.The slip strength should not be less than 95% UTS of the respective conductor.

Tension clamps shall be fitted with rivet, flat washer, cotter pin, 'U' bolts & nuts, flat washers & spring washers.

5.0 MARKING:

Tension Hardware Fittings suitable for ACSR Wolf/Dog Conductor should be supplied in a string. Tension Clamp shall be legibly & indelibly marked with Name or Trade mark of manufacturer, year of manufacture, 70 KN and suitable identification marking of the property 'WBSDEL'.

The marking should be engraved/embossed.

6.0 TESTS AND TEST CERTIFICATES :

Type Test

Along with the bid, the bidder has to submit Type Test Reports on same fittings carried out within last five years from the date of opening of techno-commercial bid of the tender from any NABL accredited Laboratory having NABL logo. Otherwise the tender may be rejected.

The following shall constitute the Type Test for Tension Clamps

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test (Slip Strength Test)
4. Ultimate Tensile Strength Test
5. Chemical Composition Test
6. Galvanizing/Electroplating Test (for ferrous parts)

The following shall constitute Type Tests on tension hardware fittings except Clamp:

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test
4. Galvanizing Test
5. Chemical Composition Test

ACCEPTANCE TEST

The following shall constitute acceptance test for Clamps:

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test (Slip Strength Test)
4. Ultimate Tensile Strength Test
5. Galvanizing Test (for ferrous parts)
6. Chemical Composition Test

Acceptance Test for tension hardware fittings except Clamp

1. Visual Examination
2. Verification of dimensions
3. Ultimate Tensile Strength Test
4. Galvanizing Test
5. Chemical Composition Test

8.0 TOLERANCE :

Tolerance shall be as per IS: 2486 (Part-I).

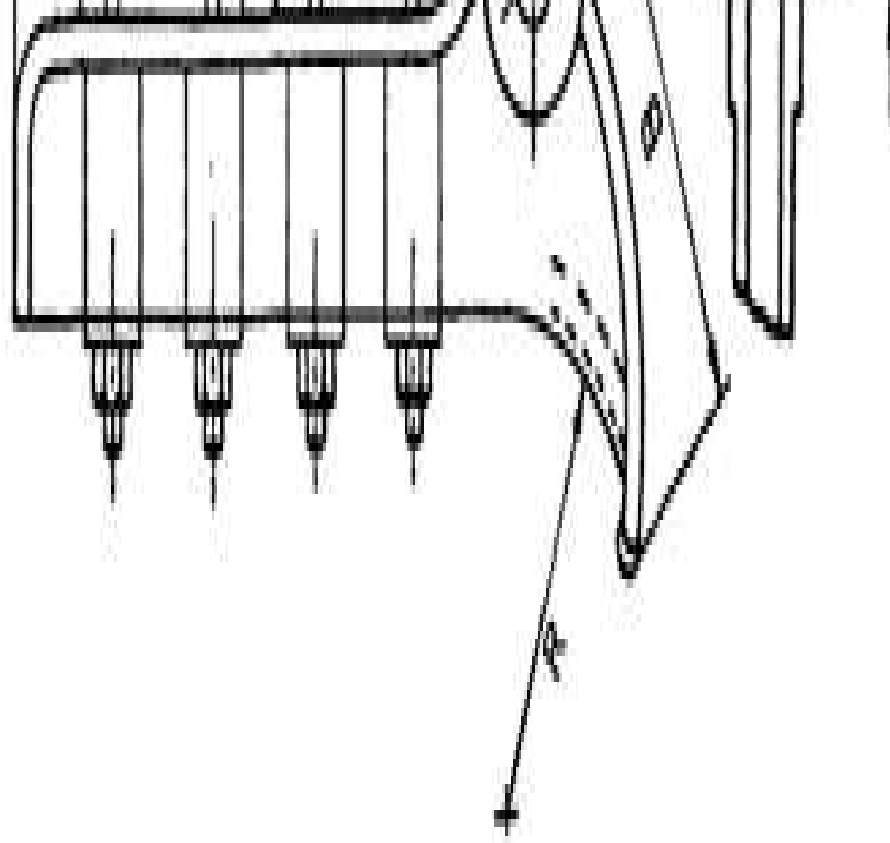
9.0 GUARANTEE:

In the event of any defect in the equipment/materials arising out of faulty design, materials, workmanship within a period of 12 (twelve) months of commissioning or 18 (eighteen) months from the date of last despatch of any integral part of the equipment/materials whichever is earlier the supplier shall guarantee to replace or repair the same to the satisfaction of the purchaser.

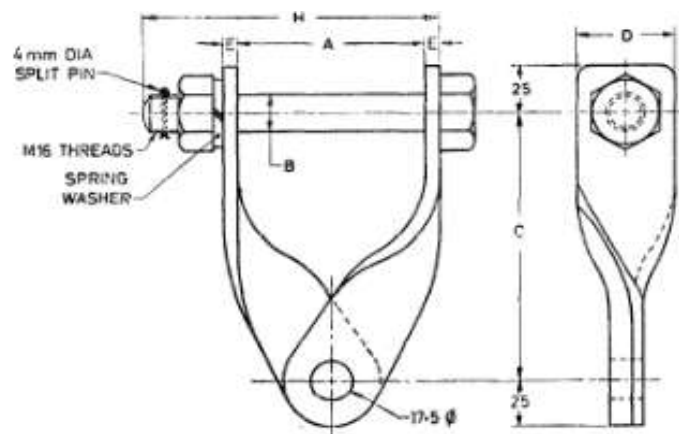
If the supplier fail to do so within a reasonable time, WBSEDCL reserves the right to effect repair or replacement by any other agency and recover charges for repair or replacement from the supplier.

GUARANTEED TECHNICAL PARTICULARS FOR TENSION HARDWARE FITTINGS FOR T & C TYPE DISC INSULATOR COMPRISING OF CROSS-ARM STRAP BOLTED TYPE TENSION CLAMP TO BE USED FOR ACSR WOLF & ACSR DOG CONDUCTOR

1.	Name & Address of the Firm	
2.	Type of Insulator Fittings	T & C type
3.	Diameter of Ball	16 mm
4.	Ultimate Tensile Strength	
a)	Bolted Type Tension Clamp	70 KN
b)	Complete Fittings	70 KN
5.	Whether dimension other than specified are in accordance to IS: 2486 (Part-II)-1989	
6.	Material used	
a)	Cross arm Strap	Mild Steel, hot dip galvanized
b)	Ball Clevis	Forged Steel and hot dipped galvanized
c)	Tongue Clevis	Forged Steel and hot dipped galvanized
d)	Bolted type Tension Clamp & keeper	High Strength Aluminium Alloy (LM-6)
g)	U-Bolts & Nut	MS, Hot Dip Galvanized
h)	Hexagonal Bolt & Nut	MS, Hot Dip Galvanized
i)	Spring Washer	Spring Steel, Electro Galvanized
j)	Plain Washer	MS Hot Dip Galvanized
k)	Split Pin, R-type Cotter pin	Stainless Steel
7	Zinc conforming to grade	As per IS 209: 1992
8	Weight and Dimension (per set)	
a)	Gross weight of complete fittings	
b)	Weight of Clamp & keeper	
c)	Weight of Cross Arm Strap	
d)	Weight of Ball Clevis	
e)	Weight of Tongue Clevis	
f)	Weight of Nut, Bolts, Rivet, U-Bolt & accessories	
g)	Weight of non ferrous parts	
h)	Minimum Guaranteed weight of Fittings	
i)	Whether galvanizing conform to IS-2633/1964	
9.	Packing details	
a)	Net Weight	
b)	Gross Weight	
c)	Contents of each Pack	
d)	Type of Packing	
9.	Tolerance in dimension, if any	IS:2486 Part I
10.	Marking	Name or Trade mark of manufacturer, year of manufacture, 70 KN and suitable identification marking of the property 'WBSEDCL', Scheme Name-IPDS on AL Clamp by engraved/embossed
11.	Guarantee	<u>12 (twelve) months of commissioning or 18 (eighteen) months from the date of last dispatch of any integral part of the equipment/materials whichever is earlier the supplier shall guarantee to replace or repair the same to the satisfaction of the purchaser.</u>

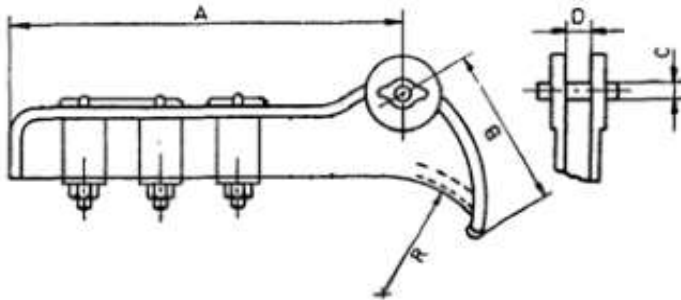


CROSS ARM STRAP (fig-2)



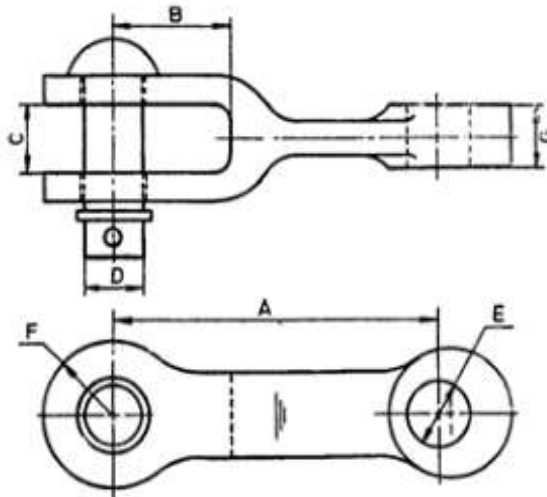
A	B (Dia)	C	D	E	H
100	16	140	35	6	145

BOLTED TYPE TENSION CLAMP (fig-3)



Material	Conductor Diameter		Dimensions					Number of U-Bolts
	Min	Max	A	B	C	D	R	
Aluminium alloy	7.6	17.8	222.2	107.6	22.2	16	76.2	3

TONGUE AND CLEVIS (fig- 4)



Dimensions, mm						
A	B	C	D	E	F	G
89	32	19	16	17.5	19	16
102	35	19	16	17.5	22	19
102	35	19	16	17.5	22	25
140	54	22	19	21	25	29
140	54	22	19	21	25	32
140	54	22	19	21	25	35
140	54	22	19	21	25	38